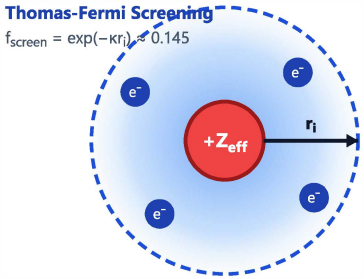
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# Unified First-Principles Mathematical Workflow: From Nuclear Ion Affinity to Glass Transition ($T_g$)

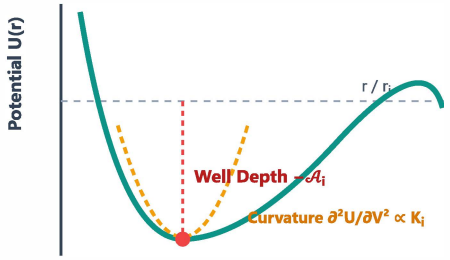
**Stage 1: Screened Electrostatics ➔ Stage 2: Rose UBER Moduli ➔ Stage 3: Topological & Charge Mismatch ➔ Stage 4: Miedema Enthalpy ➔ Stage 5: VRH Bounds ➔ Stage 6: CSM Thermal Arrest ($T_g$)**

## 1. Screened Nuclear Ion Affinity ($\mathcal{A}_i$)



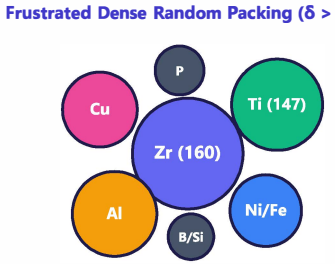

$$\mathcal{A}_i = [\,(Z_{\mathrm{eff},i} \cdot e^2) \,/\, (4\pi\,\varepsilon_0\, r_i)\,] \cdot f_{\mathrm{screen}}$$

## 2. Rose UBER Potential & Moduli ($K_i$, $G_i$)



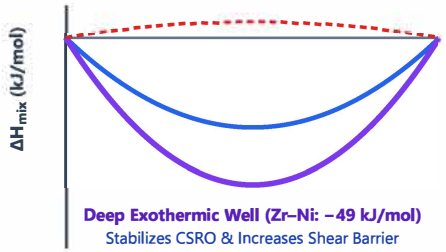

$$K_i = \mathcal{A}_i \,/\, (12\pi\,\alpha_i^{\,2}\, r_i^{\,3}), \quad G_i = K_i \cdot [3(1 - 2\nu_0)] \,/\, [2(1 + \nu_0)]$$

## 3. Topological & Electronic Mismatch (δ, Φ)



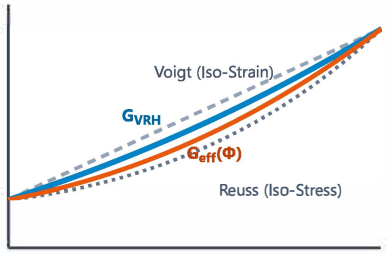

$$\delta = \sqrt{[\,\Sigma\, x_i\,(1 - r_i / \bar{r})^2\,]}, \quad \Phi = \alpha\,\sigma_r^{\,2} + \beta\,\sigma_Z^{\,2}$$

## 4. Miedema Chemical Stabilization ($\Delta H_{mix}$)



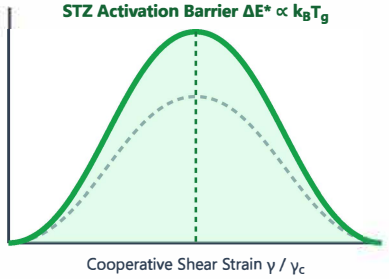

$$\Delta H_{\mathrm{mix}} = \Sigma_{i<j}\, 4 \cdot \Delta H_{ij}^{\mathrm{chem}} \cdot x_i\, x_j$$

## 5. Voigt-Reuss-Hill Composite Moduli ($K$, $G_{eff}$)




$$G_{\mathrm{VRH}} = \tfrac{1}{2}\,[\,\Sigma\, x_i G_i + (\Sigma\, x_i / G_i)^{-1}\,], \quad G_{\mathrm{eff}} = G_{\mathrm{VRH}}(1 - \tfrac{1}{2}\Phi)$$

## 6. Cooperative Shear Model (CSM) for $T_g$




$$T_g = C_{\mathrm{csm}} \cdot (G_{\mathrm{eff}} \cdot V_m \,/\, R) \cdot (1 - \Delta H_{\mathrm{mix}} \,/\, H_{\mathrm{ref}})$$